BIOCHEMISTRY CUMULATIVE EXAM

November 16, 1996

1. (35 points) Explain what is meant by each of the following terms from the literature of molecular biology. Briefly explain the importance of each listed item for biochemistry, biology or biotechnology:

a. reverse transcriptase

b. promoter

c. enhancer

d. frameshift mutation

e. nonsense mutation

f. intron (or intervening sequence)

g. restriction enzyme

2. A recent article in *Science* stated that "recognizable homologs for four of the aminoacyl-tRNA synthetases (glutamine, asparagine, lysine and cysteine) were not identified in the *M. jannaschii* genome." (The complete *M. jannaschii* genome sequence was given in the article.)

a. (5 points) What does the statement mean by "recognizable homologs?"

b. (20 points) What is surprising about the above finding?

c. (40 points) For each of the amino acids listed, <u>speculate</u> as to a <u>reasonable</u> mechanism by which the organism could possibly use the amino acid in question and live without the respective aminoacyl-tRNA synthetase. [No credit will be given for the trivial answer of saying that the organism may not need to use the amino acid in question. Try to think of other plausible answers.]